MALTEPE UNIVERSITY

Faculty of Engineering and Natural Sciences - Software Engineering

SE 342

Software Validation and Testing

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Prepared by:

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Library System

🔗 *For detailed tasks and progress, see the Jira board:*

https://librarysystem.atlassian.net/jira/software/projects/SCRUM/list

🔗 *For source code and repository structure, see the GitHub project:*

https://github.com/cerenyasarr/SE342-LibrarySystem

🔗 *For UI/UX designs and prototypes, see the Figma project:*

https://www.figma.com/design/BbQ4pCkzTqrPuusHIJ0hLw/Attendance-System?node-id=0-1&t=WzjFoxee5FbCb83I-1

1. Introduction

1.1 Project Purpose

The primary purpose of this project is to design and develop a web-based **Library Management System** to streamline and automate the core operations of a library. Currently, manual methods of tracking books and borrowing records are inefficient and prone to errors. This system aim to provide a centralized platform where **Administrators** can easily manage book inventory and user records, while **Students** can efficiently search for resources, borrow books, and track their reading history. Ultimately, the project seeks to enhance the overall user experience and operational efficiency of the library environment.

1.2 Scope

The scope of this project encompasses the development of a functional software application that covers the following key modules:

- **Authentication & Authorization:** Secure login and registration systems with distinct roles for Administrators (librarians) and Users (students).
- **Book Inventory Management:** A dynamic catalog system that allows for adding, updating, and deleting book records, as well as tracking real-time stock availability.
- **Search & Discovery:** A user-friendly interface enabling users to search for books by title, author, or category with advanced filtering options.

- **Circulation Management:** A transaction system to handle book borrowing and returning processes, including due date tracking and status updates.
- **User Dashboard:** A personalized area for users to view their active loans, borrowing history, and profile details.

2. Requirements

2.1 Functional Requirements

1. **Book Search:**

   Members shall be able to search for books by title, author, ISBN, or category.

2. **Advanced Filtering:**

   The system shall allow filtering by availability (available, reserved, borrowed).

3. **Book Reservation:**

   Members shall be able to reserve a book when no copies are available.

4. **Cancel Reservation:**

   Members shall be able to cancel their pending reservations.

5. **Book Borrowing:**

   Members shall be able to borrow books that are currently available.

6. **Duplicate Borrowing Prevention:**

   The system shall prevent borrowing the same physical copy more than once simultaneously.

7. **Librarian Approval:**

   Librarians shall be able to approve or reject borrowing and reservation requests.

8. **Automatic Availability Update:**

   The system shall automatically update stock when a book is borrowed, returned, or reserved.

9. **Borrowing History:**

   Members shall be able to view their active and past borrowings.

10. **Due Date Calculation:**

    The system shall calculate the return date automatically based on borrowing rules.

11. **Overdue Tracking:**

    The system shall identify overdue books and notify the librarian or member.

12. **Member Notifications:**

    Members shall receive alerts for reservation approvals, due dates, and overdue items.

2.2 Non-Functional Requirements

1. **Performance:**

   System response time shall be under **3 seconds** for all major interactions.

2. **Security:**

   The system shall enforce secure login, authentication, and role-based authorization.

3. **Data Integrity:**

   All operations shall maintain accurate and consistent data across the system.

4. **Availability:**

   The system shall maintain at least **99% uptime** during operational hours.

5. **Usability:**

   The interface shall be intuitive, easy to navigate, and accessible for both librarians and members.

6. **Compatibility:**

   The system shall fully support major browsers (Chrome, Safari, Firefox, Edge) and desktop screen resolutions.

7. **Scalability:**

   The system shall be capable of supporting an increasing number of users and books without performance degradation.

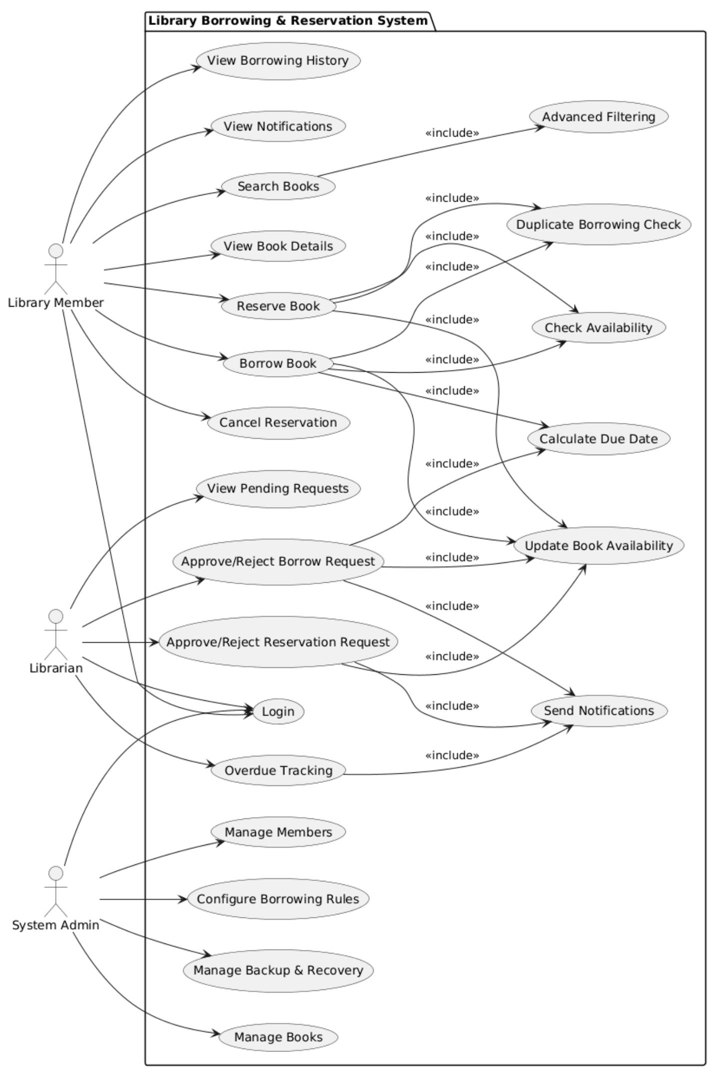
8. **Backup & Recovery:**

   The system shall perform automatic daily backups and support quick data restoration in case of failure.

3. System Design

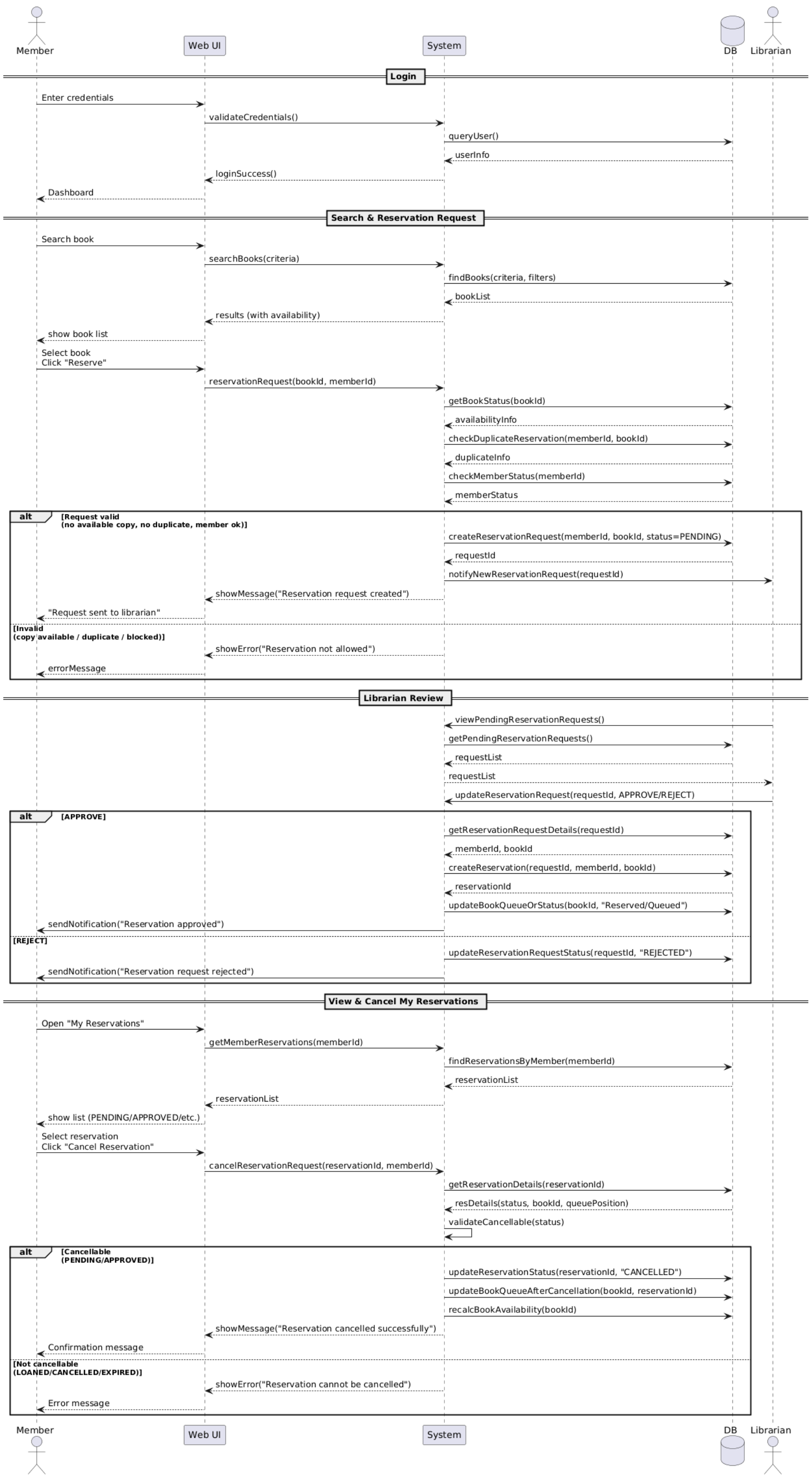
3.1 UML Diagrams

3.1.1 Use Case Diagram

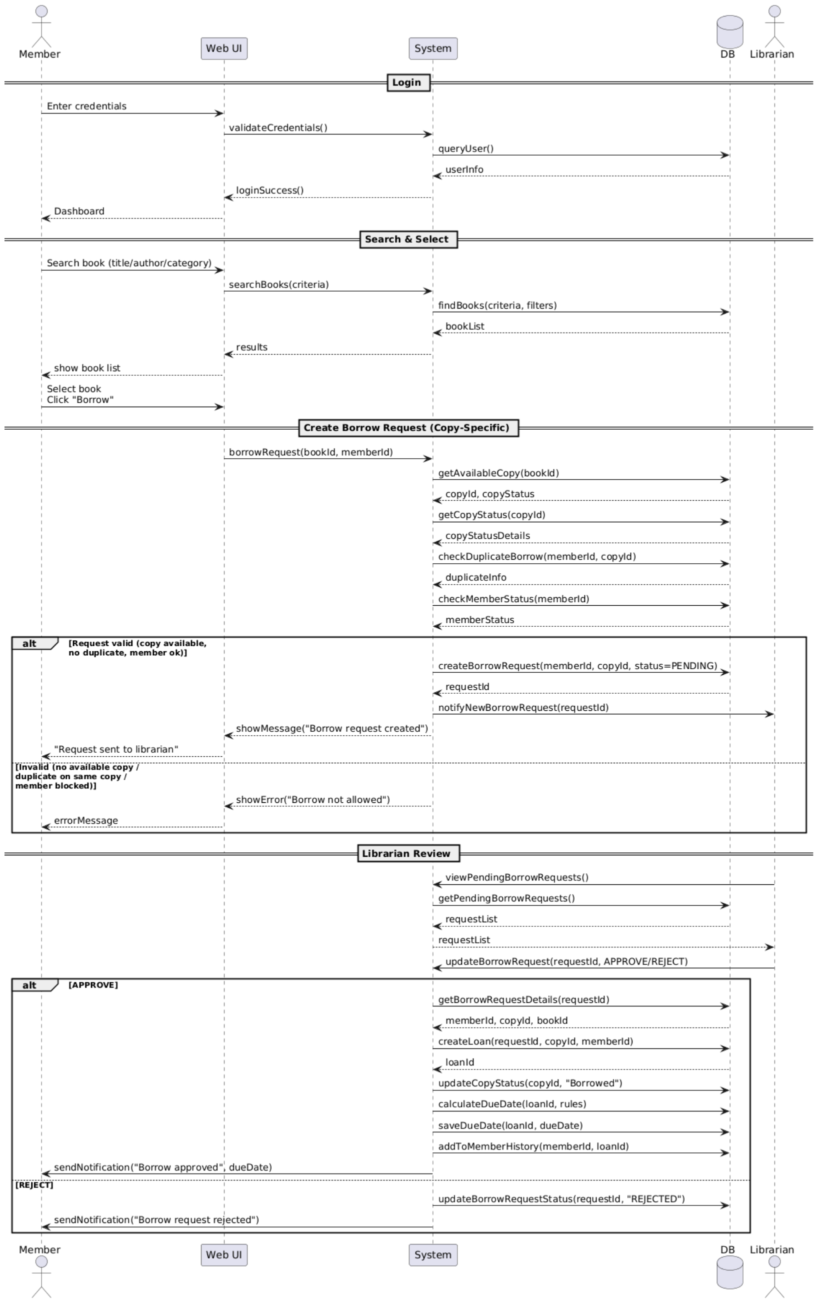


3.1.2 Sequence Diagrams

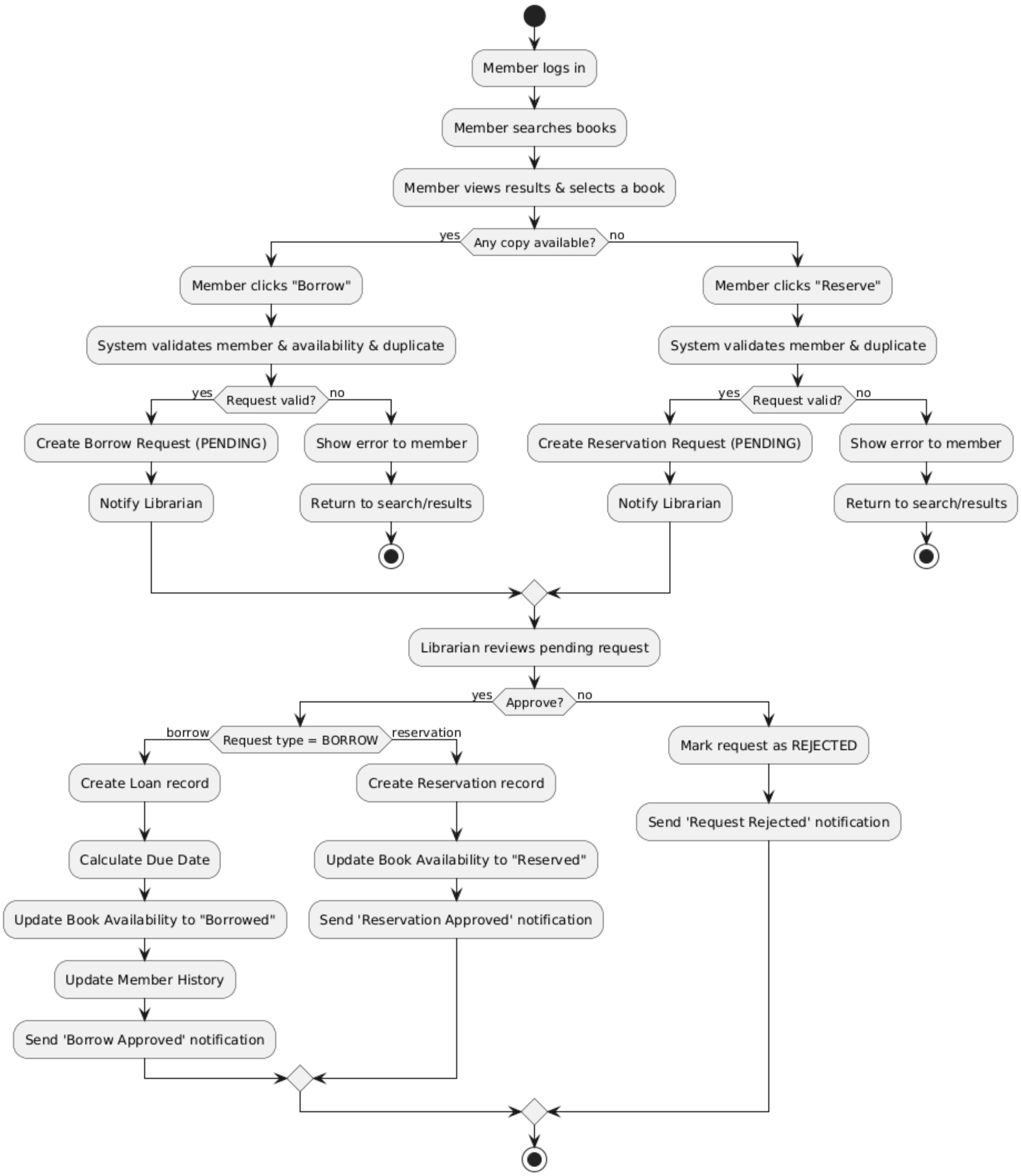
- Reservation Workflow



- ## Borrowing Workflow

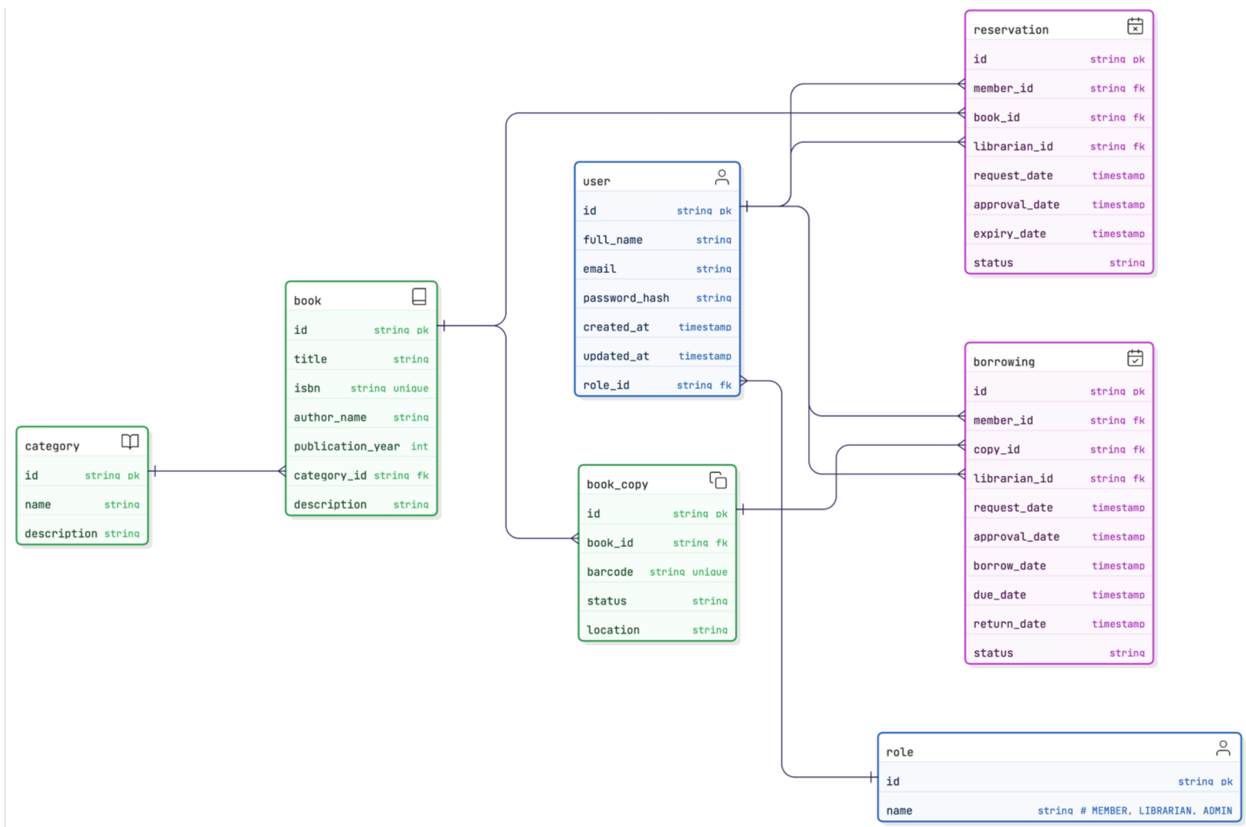


3.1.3 BONUS: ACTIVITY DIAGRAM



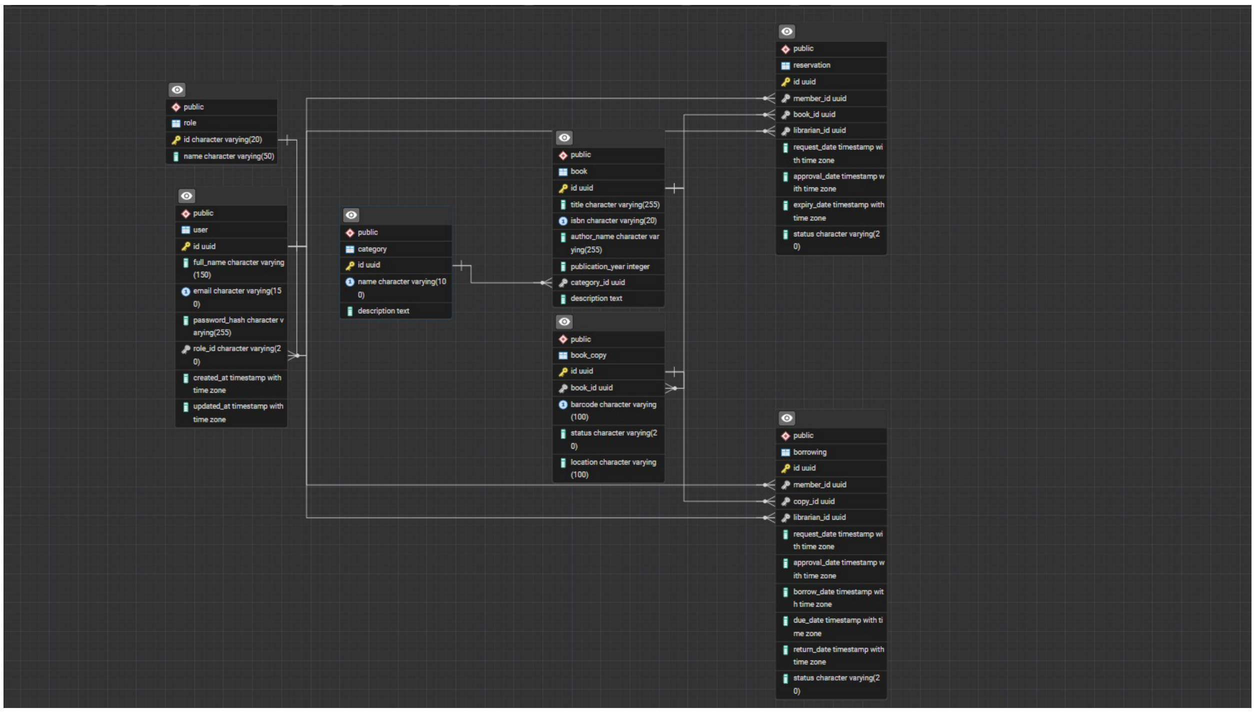
3.2 ERD (Entity Relationship Diagram)

The ERD was designed using Crow's Foot notation to clearly represent system entities, attributes, and relationships. Core entities were derived directly from the functional requirements: User, Role, Category, Book, BookCopy, Borrowing, and Reservation. Roles differentiate system actors—Member, Librarian, and Administrator—without requiring separate user tables. One-to-many relationships ensure logical data flow, such as Category → Book and Book → BookCopy. Transactional processes like borrowing and reservations are modeled as independent tables to preserve history and support approval workflows. The ERD ensures data consistency, prevents duplicate borrowing of the same copy, and supports future system extensions.



3.3 Database Schema Design (PostgreSQL)

The logical ERD was implemented as a normalized PostgreSQL relational database. Each table includes primary keys, foreign keys, NOT NULL constraints, unique fields, and timestamps to maintain referential integrity and traceability. UUID values were chosen for identifiers to ensure global uniqueness and scalability. Business rules—such as unique ISBNs, copy barcodes, valid reservation approvals, and enforced role assignments—are implemented at schema level through constraints. Dummy data was inserted to validate relationships, system behavior, and common user scenarios. The resulting schema is structured, maintainable, and ready for integration with backend services and API development.



- TABLE:

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";

-- 1) ROLE
CREATE TABLE role (
   id      VARCHAR(20) PRIMARY KEY,   -- örn: 'MEMBER', 'LIBRARIAN', 'ADMIN'
   name    VARCHAR(50) NOT NULL
);

-- 2) USER
CREATE TABLE "user" (
```

```
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  full_name       VARCHAR(150) NOT NULL,
  email          VARCHAR(150) NOT NULL UNIQUE,
  password_hash   VARCHAR(255) NOT NULL,
  role_id        VARCHAR(20) NOT NULL,
  created_at      TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_at      TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  CONSTRAINT fk_user_role
    FOREIGN KEY (role_id) REFERENCES role(id)
);

-- 3) CATEGORY
CREATE TABLE category (
  id         UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name       VARCHAR(100) NOT NULL UNIQUE,
  description TEXT
);

-- 4) BOOK
CREATE TABLE book (
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  title             VARCHAR(255) NOT NULL,
  isbn              VARCHAR(20) UNIQUE,
  author_name        VARCHAR(255),
  publication_year    INT,
  category_id        UUID NOT NULL,
  description        TEXT,
  CONSTRAINT fk_book_category
    FOREIGN KEY (category_id) REFERENCES category(id)
);

-- 5) BOOK_COPY
CREATE TABLE book_copy (
  id         UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  book_id     UUID NOT NULL,
  barcode     VARCHAR(100) NOT NULL UNIQUE,
  status      VARCHAR(20) NOT NULL,   -- AVAILABLE / BORROWED / RESERVED /
LOST (uygulama tarafında)
  location    VARCHAR(100),
  CONSTRAINT fk_copy_book
    FOREIGN KEY (book_id) REFERENCES book(id)
);

-- 6) BORROWING
CREATE TABLE borrowing (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    member_id       UUID NOT NULL,
    copy_id         UUID NOT NULL,
    librarian_id    UUID,              -- isteği sonradan onaylayacak kişi, o yüzden NULL olabilir
    request_date    TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    approval_date   TIMESTAMPTZ,
    borrow_date     TIMESTAMPTZ,
    due_date        TIMESTAMPTZ,
    return_date     TIMESTAMPTZ,
    status          VARCHAR(20) NOT NULL,   -- PENDING / APPROVED / REJECTED /
BORROWED / RETURNED / OVERDUE
    CONSTRAINT fk_borrow_member
        FOREIGN KEY (member_id) REFERENCES "user"(id),
    CONSTRAINT fk_borrow_copy
        FOREIGN KEY (copy_id) REFERENCES book_copy(id),
    CONSTRAINT fk_borrow_librarian
        FOREIGN KEY (librarian_id) REFERENCES "user"(id)
);

-- 7) RESERVATION
CREATE TABLE reservation (
    id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    member_id       UUID NOT NULL,
    book_id         UUID NOT NULL,
    librarian_id    UUID,              -- onaylanana kadar boş olabilir
    request_date    TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    approval_date   TIMESTAMPTZ,
    expiry_date     TIMESTAMPTZ,
    status          VARCHAR(20) NOT NULL,   -- PENDING / APPROVED / REJECTED /
CANCELLED / FULFILLED
    CONSTRAINT fk_res_member
        FOREIGN KEY (member_id) REFERENCES "user"(id),
    CONSTRAINT fk_res_book
        FOREIGN KEY (book_id) REFERENCES book(id),
    CONSTRAINT fk_res_librarian
        FOREIGN KEY (librarian_id) REFERENCES "user"(id)
);
```

- Dummy Datas:

```
BEGIN;

-- ROLE
INSERT INTO role (id, name) VALUES
```

```
('ADMIN',     'System Administrator'),
('LIBRARIAN', 'Librarian'),
('MEMBER',    'Library Member');

-- USER
INSERT INTO "user" (id, full_name, email, password_hash, role_id, created_at, updated_at)
VALUES
('8f7d9b63-0f9b-4b9b-9f5e-8c1a2b5f1a11', 'Alice Admin',       'admin@library.com',
'hashed_admin_pass',   'ADMIN',     NOW(), NOW()),
('b3c2e4d7-1a2b-4c5d-8e9f-112233445566', 'Laura Librarian',   'laura@library.com',
'hashed_librarian_pw', 'LIBRARIAN', NOW(), NOW()),
('c1d2e3f4-5a6b-4c7d-8e9f-223344556677', 'John Miller',       'john.miller@example.com',
'hashed_member1_pw',   'MEMBER',    NOW(), NOW()),
('d2e3f4a5-6b7c-4d8e-9f10-334455667788', 'Sarah Carter',
'sarah.carter@example.com','hashed_member2_pw',   'MEMBER',    NOW(), NOW()),
('e3f4a5b6-7c8d-4e9f-a011-445566778899', 'David Wilson',
'david.wilson@example.com','hashed_member3_pw',   'MEMBER',    NOW(), NOW());

-- CATEGORY
INSERT INTO category (id, name, description) VALUES
('11111111-aaaa-4aaa-8aaa-aaaaaaaaaaaa', 'Science Fiction',   'Futuristic, speculative fiction'),
('22222222-bbbb-4bbb-8bbb-bbbbbbbbbbbb', 'Computer Science',  'Programming, software
engineering, algorithms'),
('33333333-cccc-4ccc-8ccc-cccccccccccc', 'History',           'Historical studies and research'),
('44444444-dddd-4ddd-8ddd-dddddddddddd', 'Fantasy',           'Fantasy worlds, magic, epic
adventures'),
('55555555-eeee-4eee-8eee-eeeeeeeeeeee', 'Classic Literature','Canonized classic novels');

-- BOOK
INSERT INTO book (id, title, isbn, author_name, publication_year, category_id, description)
VALUES
('aaaa1111-0000-4000-8000-000000000001',
 'Dune',
 '9780441172719',
 'Frank Herbert',
 1965,
 '11111111-aaaa-4aaa-8aaa-aaaaaaaaaaaa',
 'Classic science fiction novel set on the desert planet Arrakis.'),

('aaaa1111-0000-4000-8000-000000000002',
 'Clean Code: A Handbook of Agile Software Craftsmanship',
 '9780132350884',
 'Robert C. Martin',
 2008,
 '22222222-bbbb-4bbb-8bbb-bbbbbbbbbbbb',
 'Best practices for writing clean, maintainable code.'),
```

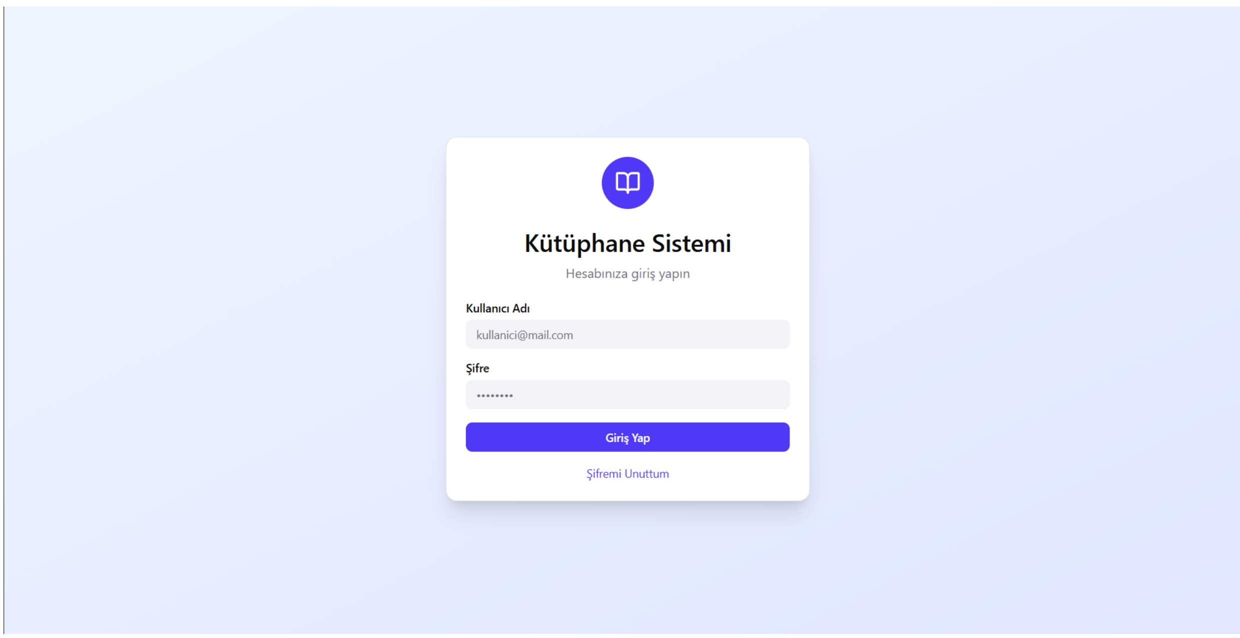
```
('aaaa1111-0000-4000-8000-000000000003',
 'Introduction to Algorithms',
 '9780262033848',
 'Thomas H. Cormen',
 2009,
 '22222222-bbbb-4bbb-8bbb-bbbbbbbbbbbb',
 'Comprehensive textbook on algorithms.'),

('aaaa1111-0000-4000-8000-000000000004',
 'Sapiens: A Brief History of Humankind',
 '9780062316110',
 'Yuval Noah Harari',
 2011,
 '33333333-cccc-4ccc-8ccc-cccccccccccc',
 'A narrative of human history from prehistoric times to the present.'),

('aaaa1111-0000-4000-8000-000000000005',
 'The Fellowship of the Ring',
 '9780261102354',
 'J.R.R. Tolkien',
 1954,
 '44444444-dddd-4ddd-8ddd-dddddddddddd',
 'First volume of The Lord of the Rings epic fantasy trilogy.');

-- BOOK_COPY
INSERT INTO book_copy (id, book_id, barcode, status, location) VALUES
-- Dune: 2 kopya
('bbbb2222-0000-4000-8000-000000000001',
 'aaaa1111-0000-4000-8000-000000000001',
 'BC-9780441172719-1',
 'AVAILABLE',
 'Shelf A1'),
('bbbb2222-0000-4000-8000-000000000002',
 'aaaa1111-0000-4000-8000-000000000001',
 'BC-9780441172719-2',
 'AVAILABLE',
 'Shelf A1'),

-- Clean Code: 2 kopya
('bbbb2222-0000-4000-8000-000000000003',
 'aaaa1111-0000-4000-8000-000000000002',
 'BC-9780132350884-1',
 'BORROWED',
 'Shelf B3'),
('bbbb2222-0000-4000-8000-000000000004',
```

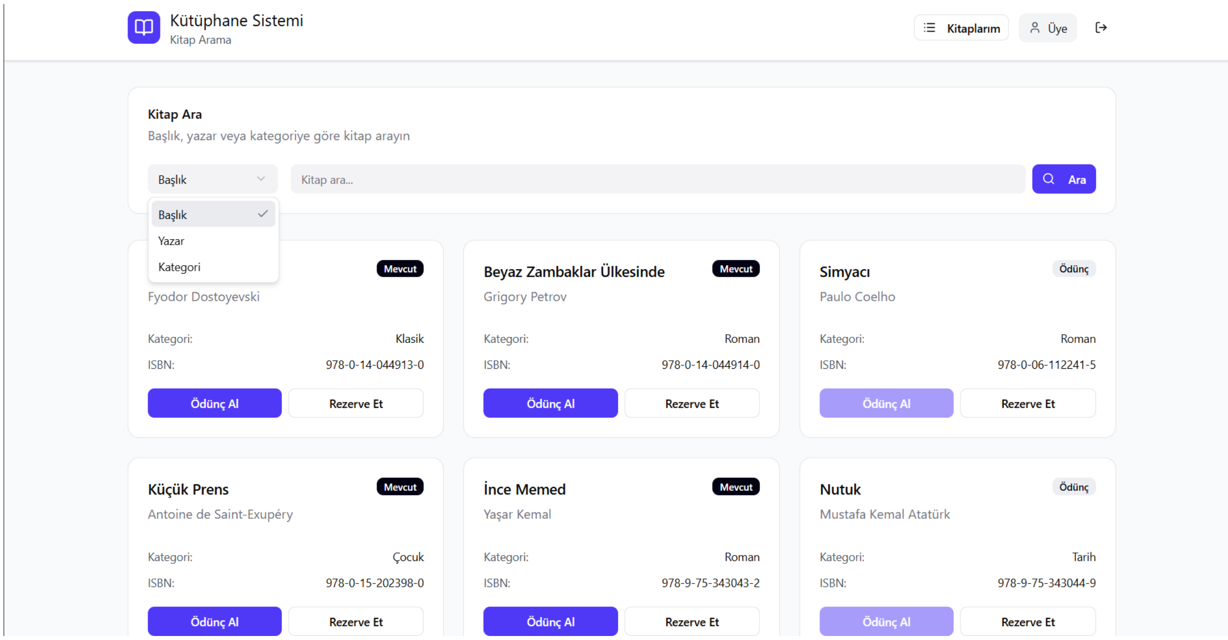
```
 'aaaa1111-0000-4000-8000-000000000002',
 'BC-9780132350884-2',
 'AVAILABLE',
 'Shelf B3'),

-- Sapiens: 1 kopya
('bbbb2222-0000-4000-8000-000000000005',
 'aaaa1111-0000-4000-8000-000000000004',
 'BC-9780062316110-1',
 'AVAILABLE',
 'Shelf C2'),

-- Fellowship of the Ring: 1 kopya
('bbbb2222-0000-4000-8000-000000000006',
 'aaaa1111-0000-4000-8000-000000000005',
 'BC-9780261102354-1',
 'AVAILABLE',
 'Shelf D4');

-- BORROWING
INSERT INTO borrowing (
    id, member_id, copy_id, librarian_id,
    request_date, approval_date, borrow_date, due_date, return_date, status
)
VALUES
-- John Miller, Clean Code kopya-1'i ödünç almış, hâlâ iade etmemiş (BORROWED)
('cccc3333-0000-4000-8000-000000000001',
 'c1d2e3f4-5a6b-4c7d-8e9f-223344556677',  -- John
 'bbbb2222-0000-4000-8000-000000000003',  -- Clean Code copy 1
 'b3c2e4d7-1a2b-4c5d-8e9f-112233445566',  -- Laura (librarian)
 NOW() - INTERVAL '7 days',
 NOW() - INTERVAL '7 days',
 NOW() - INTERVAL '7 days',
 NOW() + INTERVAL '7 days',
 NULL,
 'BORROWED'),

-- Sarah Carter, Sapiens'i almış ve geri getirmiş (RETURNED)
('cccc3333-0000-4000-8000-000000000002',
 'd2e3f4a5-6b7c-4d8e-9f10-334455667788',  -- Sarah
 'bbbb2222-0000-4000-8000-000000000005',  -- Sapiens copy 1
 'b3c2e4d7-1a2b-4c5d-8e9f-112233445566',  -- Laura
 NOW() - INTERVAL '40 days',
 NOW() - INTERVAL '39 days',
 NOW() - INTERVAL '39 days',
 NOW() - INTERVAL '25 days',
```

```
 NOW() - INTERVAL '20 days',
 'RETURNED');

-- RESERVATION
INSERT INTO reservation (
    id, member_id, book_id, librarian_id,
    request_date, approval_date, expiry_date, status
)
VALUES
-- David, Dune için rezervasyon istemiş, hâlâ bekliyor (PENDING)
('dddd4444-0000-4000-8000-000000000001',
 'e3f4a5b6-7c8d-4e9f-a011-445566778899',   -- David
 'aaaa1111-0000-4000-8000-000000000001',   -- Dune
 NULL,
 NOW() - INTERVAL '1 day',
 NULL,
 NOW() + INTERVAL '6 days',
 'PENDING'),

-- John'un Fellowship of the Ring için rezervasyonu onaylanmış (APPROVED)
('dddd4444-0000-4000-8000-000000000002',
 'c1d2e3f4-5a6b-4c7d-8e9f-223344556677',   -- John
 'aaaa1111-0000-4000-8000-000000000005',   -- Fellowship of the Ring
 'b3c2e4d7-1a2b-4c5d-8e9f-112233445566',   -- Laura (librarian)
 NOW() - INTERVAL '3 days',
 NOW() - INTERVAL '2 days',
 NOW() + INTERVAL '5 days',
 'APPROVED');

COMMIT;
```

4. User Interface (UI) Design

- Web Version:

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Kitap Arama
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Üye
Kitap Ara
Başlık, yazar veya kategoriye göre kitap arayın
Başlık
Kitap ara...
Ara
Başlık
Yazar
Kategori
Mevcut
Fyodor Dostoyevski
Kategori: Klasik
ISBN: 978-0-14-044913-0
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Grigory Petrov
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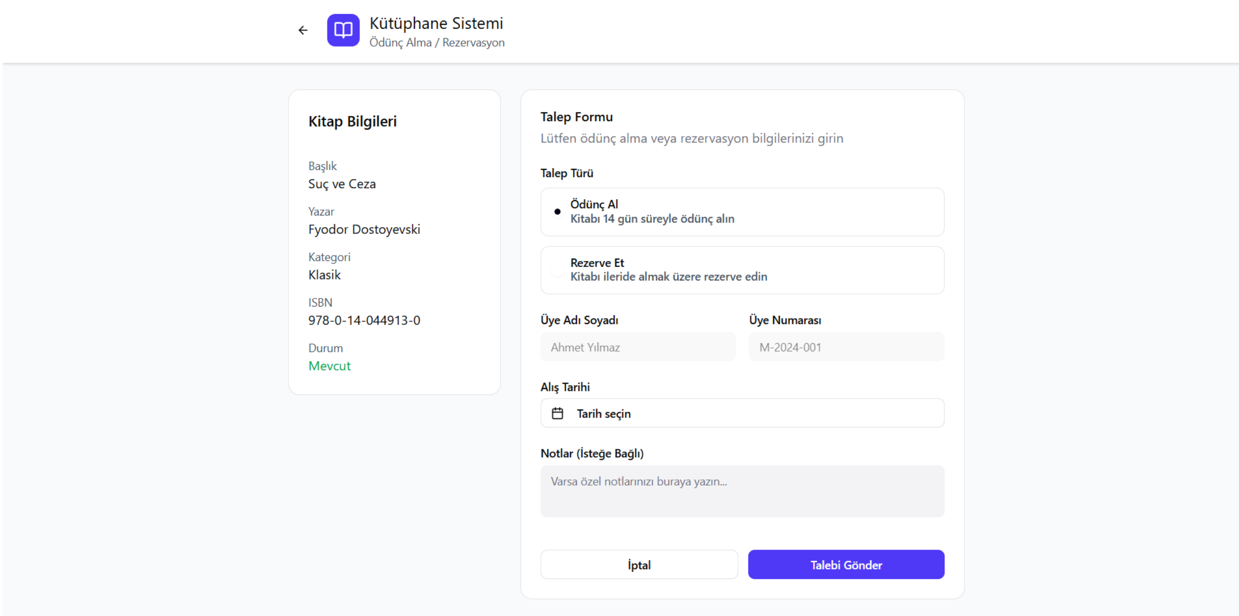
Rezerve Et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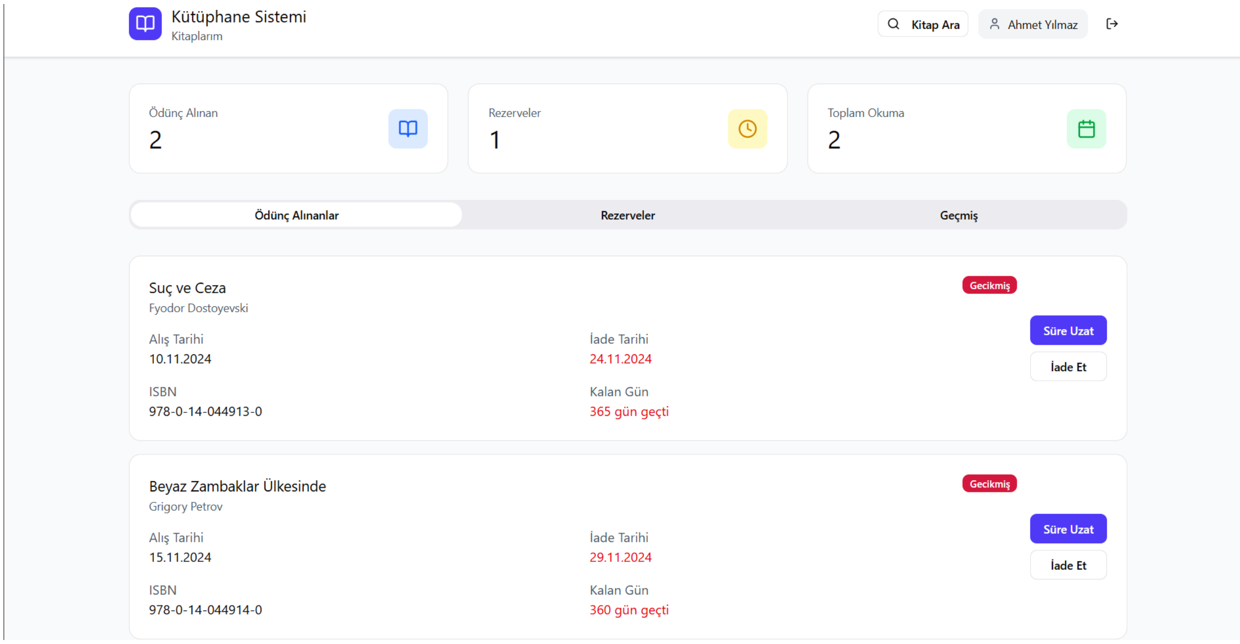
Durum
Mevcut
Talep Formu
Lütfen ödünç alma veya rezervasyon bilgilerinizi girin
Talep Türü
Ödünç Al
Kitabı 14 gün süreyle ödünç alın
Rezerve Et
Kitabı ileride almak üzere rezerve edin
Üye Adı Soyadı
Ahmet Yılmaz
Üye Numarası
M-2024-001
Alış Tarihi
Tarih seçin
Notlar (İsteğe Bağlı)
Varsa özel notlarınızı buraya yazın...
İptal
Talebi Gönder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Kitaplarım
Kitap Ara
Ahmet Yılmaz
Ödünç Alınan
2
Rezerveler
1
Toplam Okuma
2
Ödünç Alınanlar
Rezerveler
Geçmiş
Suç ve Ceza
Fyodor Dostoyevski
Gecikmiş
Alış Tarihi
10.11.2024
İade Tarihi
24.11.2024
Süre Uzat
İade Et
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Kalan Gün
365 gün geçti
Beyaz Zambaklar Ülkesinde
Grigory Petrov
Gecikmiş
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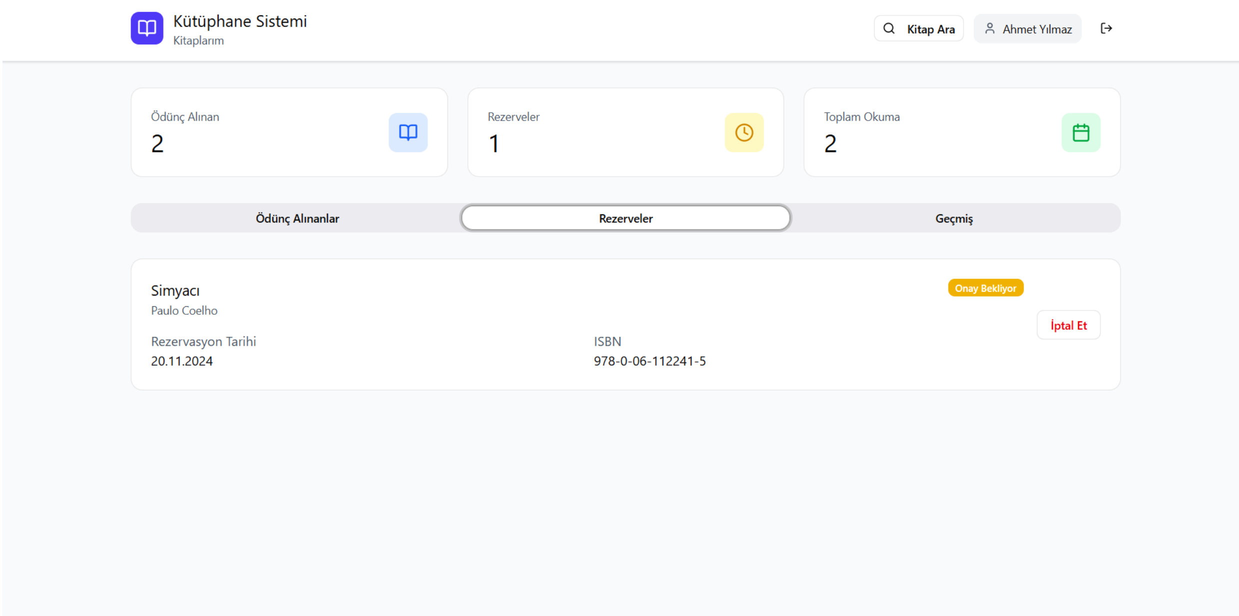
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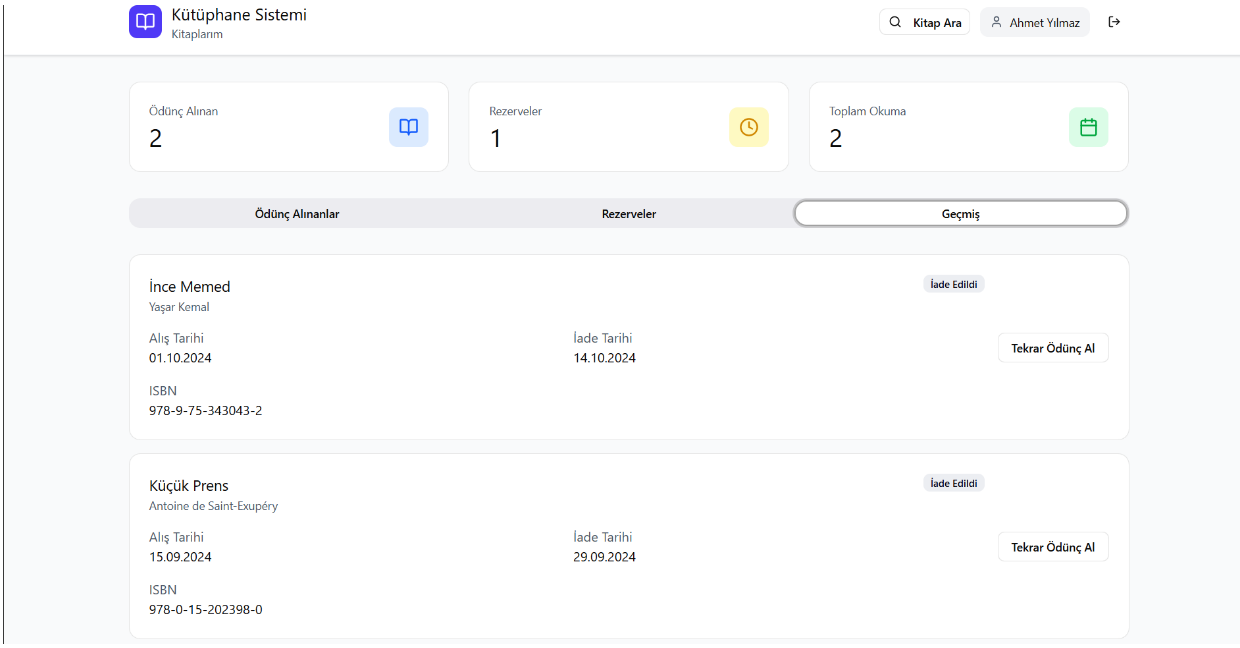

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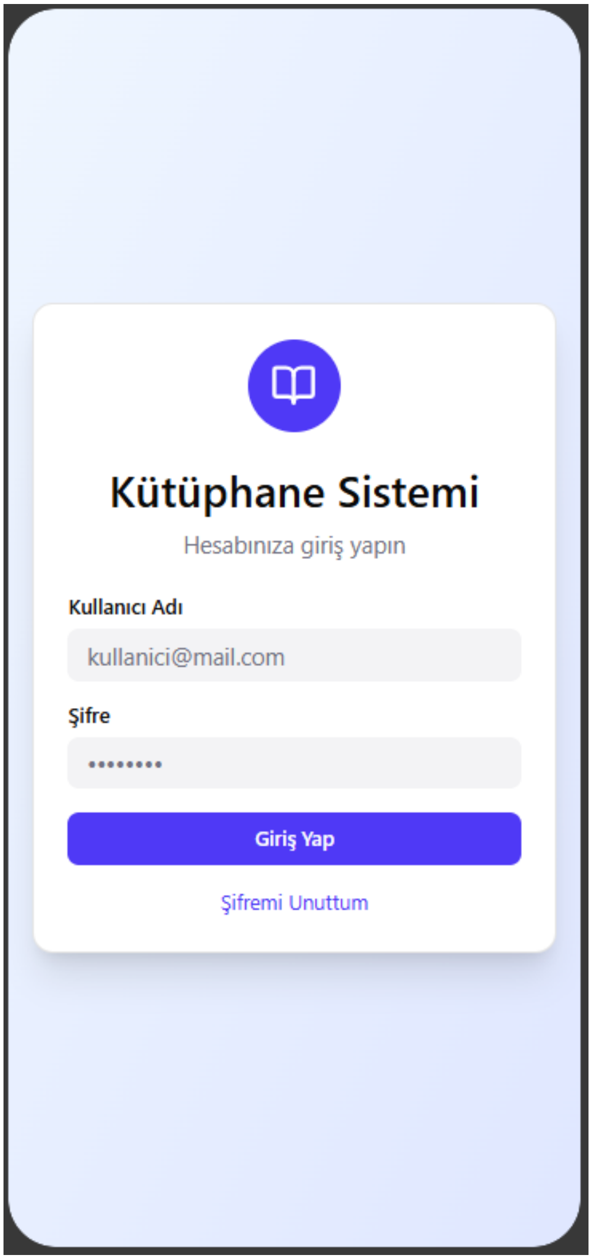
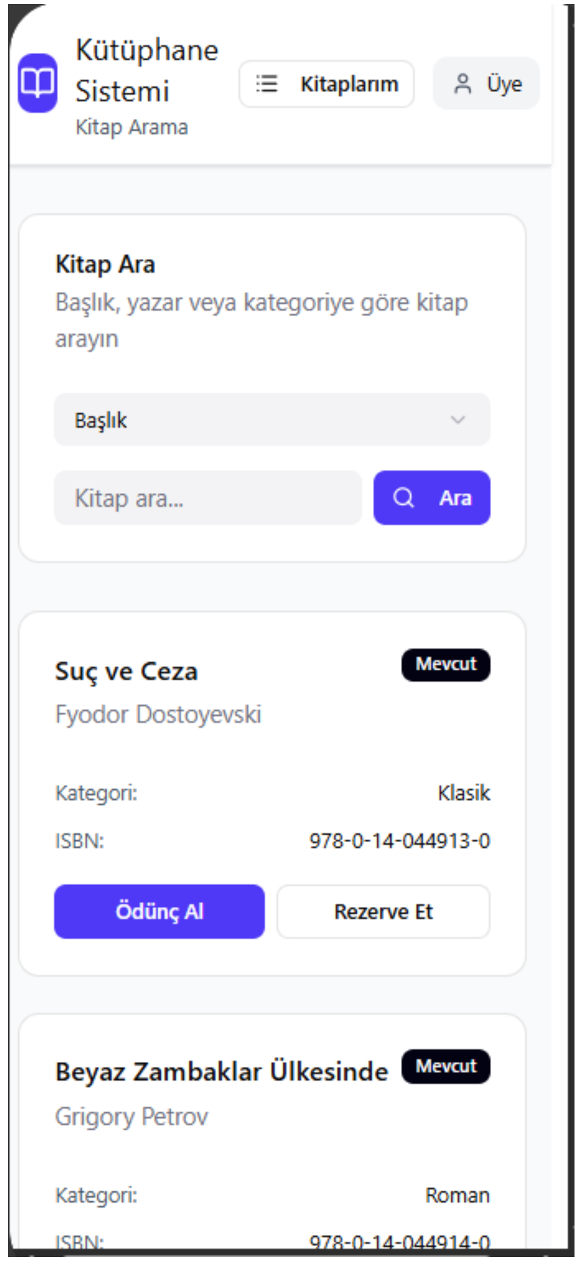
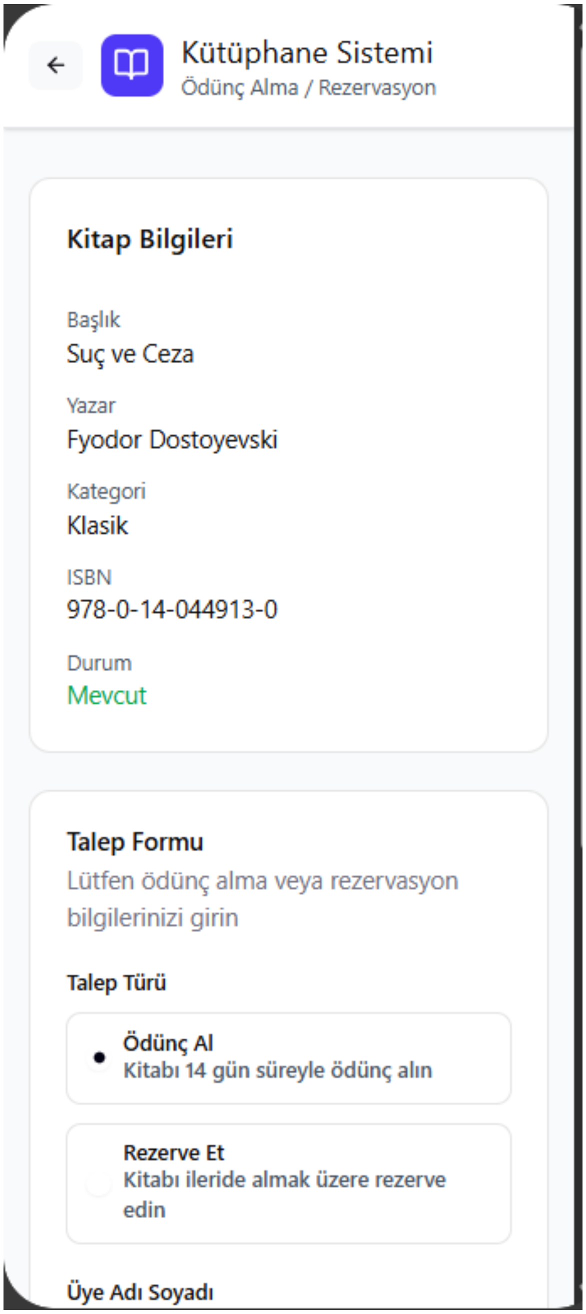
- Mobil Version:



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Talep Türü
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Rezerve Et
Kitabı ileride almak üzere rezerve edin
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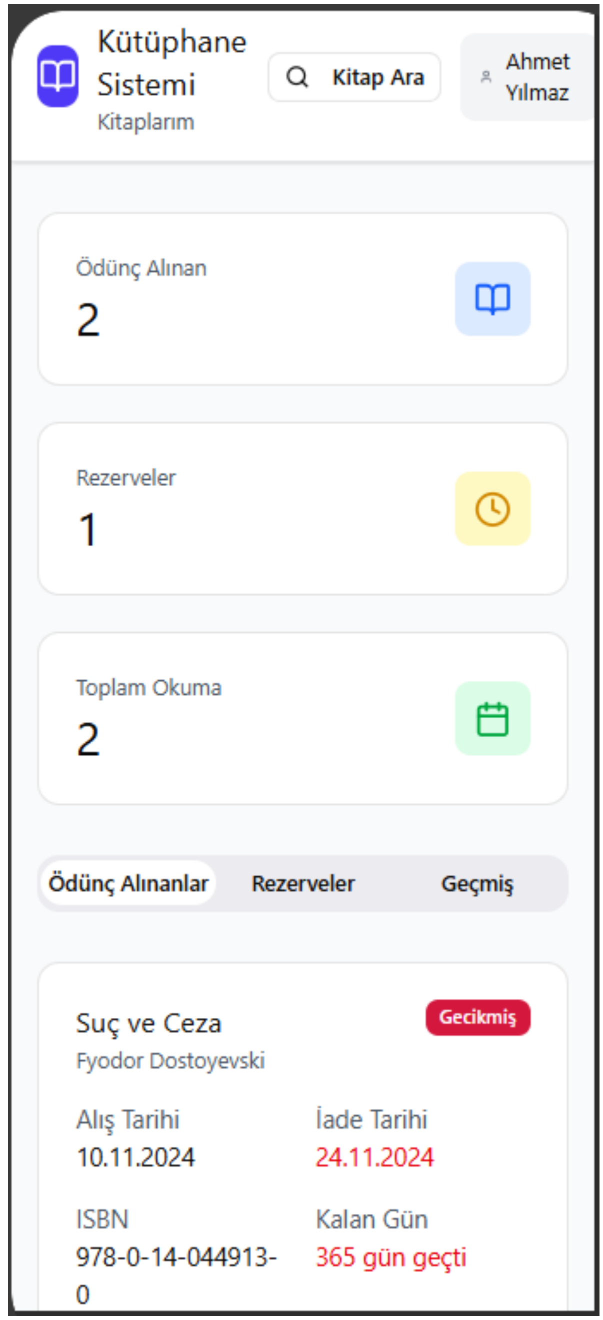
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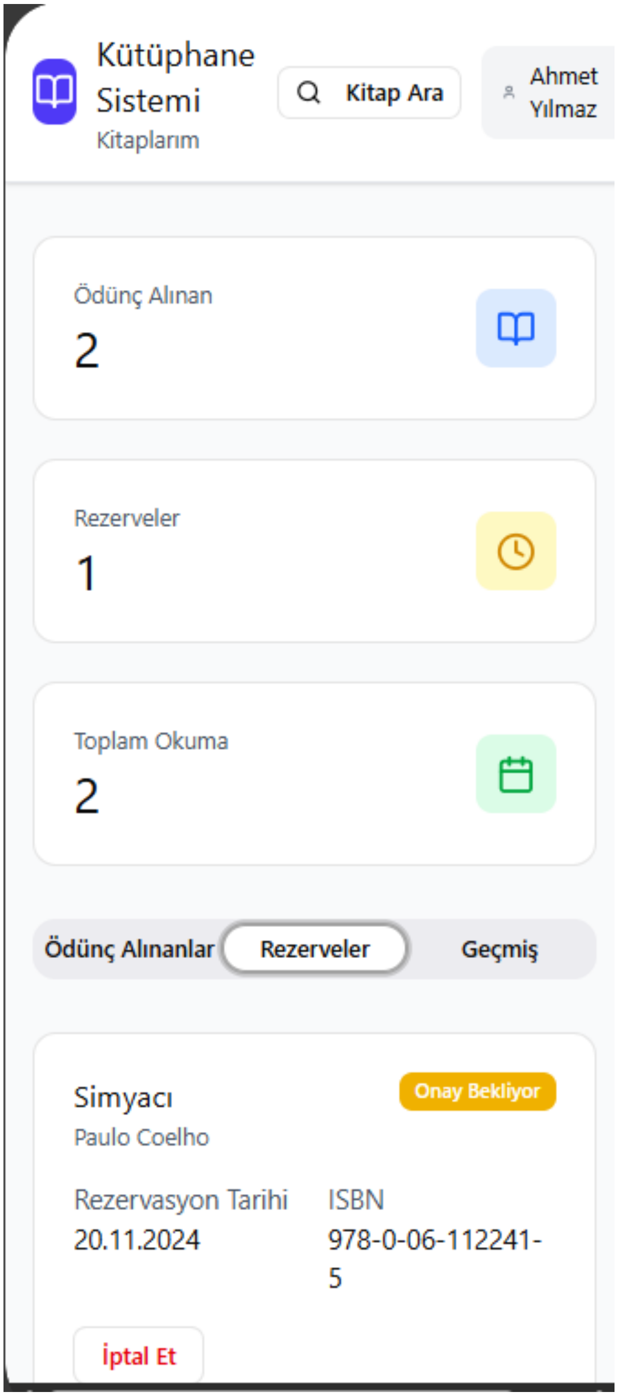
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5. Project Management & Methodology

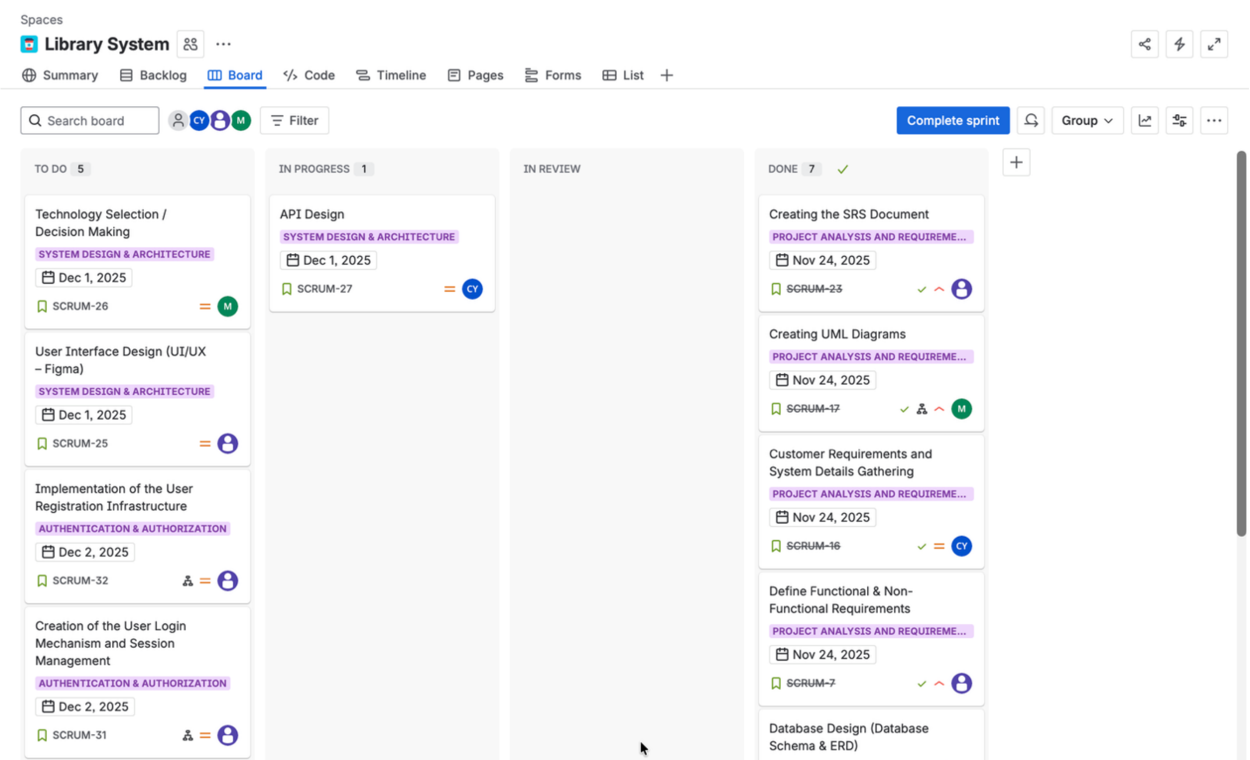
In this project, **Agile** principles have been adopted throughout the Software Development Life Cycle (SDLC), utilizing the **Scrum** framework. The development process is organized into specific time-boxed iterations (sprints) to ensure a manageable, flexible, and transparent workflow.

5.1. Project Tracking with Jira

Jira is used for project management, task assignment, and process tracking. A Kanban/Scrum board allows for transparent monitoring of the workflow.

- **Issue Types:**
  - **Epic:** Major modules of the project (e.g., "Book Discovery Module", "Authentication").
  - **User Story:** Feature requests written from the end-user's perspective.
  - **Task:** Technical setups and configurations.

Here is an image from the Jira board:



5.2 Sprint Planning

The project development process is divided into 6 sprints, including the initialization phase (Sprint 0). The focus area and expected deliverables for each sprint are outlined in the table below:

Sprint	Focus Area	Scope and Deliverables
Sprint 0	**Project Analysis & Setup**	• Completion of requirements analysis. • Creation of architectural design and database schema (ERD). • Repository initialization and project skeleton setup. • Definition of the technology stack.
Sprint 1	**Auth & Authorization**	• Implementation of Register and Login interfaces. • JWT (JSON Web Token) based authentication infrastructure. • Role-based authorization (Admin/User) controls.
Sprint 2	**Book Discovery Module**	• Creation of book listing interfaces. • Search and filtering functions (by category, author, or name). • Development of the Book Detail page (description, stock status, etc.).
Sprint 3	**Borrowing & Reservation**	• Implementation of the "Borrow" business logic. • Stock control and Reservation system. • Backend logic for due date calculation and penalty/overdue handling.
Sprint 4	**User Dashboard**	• User profile viewing and editing.

		• Display of "Current Borrows" and "Transaction History" lists. • "Favorite Books" or reading list features.
Sprint 5	**Testing & Deployment**	• Execution of Unit and Integration tests. • Deployment of the application to the production environment. • User Acceptance Testing (UAT) and bug fixing.

5.3 Version Control System

Git is used for source code management and version control, with **GitHub** serving as the remote repository. To prevent code conflicts during collaboration and to maintain a clean history, a **Feature Branch** workflow has been adopted.

- **Branching Strategy:**
  - main: The stable branch containing code ready for production.
  - develop: The integration branch where development code merges and is tested.
  - feature/*: Branches created for specific features (e.g., feature/login-screen, feature/book-api).